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TShark / Wireshark Cheat Sheet

A quick-reference guide for Wireshark and TShark: capture vs display filters, comparison and logical operators, keyboard shortcuts, and common filtering commands.

Filter Types & Modes 1

TYPE	DESCRIPTION
Capture Filter	Filters packets *during* capture (BPF syntax).
Display Filter	Hides packets from the capture display only.
Promiscuous Mode	Captures all packets on the network segment.
Monitor Mode	Wireless interface captures all traffic it can receive (Unix/Linux only).

Protocol values — `ether`, `fddi`, `ip`, `arp`, `rarp`, `decnet`, `lat`, `sca`, `moprc`, `mopdl`, `tcp`, `udp`.

Capture Filter Syntax 2

FILTER	EXAMPLE
Single host	`host 10.10.50.1`
By port	`tcp port 80`
Source host	`src host 10.0.0.1`
Destination network	`dst net 192.168.1.0/24`
Exclude a port	`not port 22`
Port range	`tcp portrange 6000-6010`
Combined	`tcp and dst port 443`

Syntax: `protocol direction hosts value logical-operator expression`

Example: `tcp src 192.168.1.1 80 and tcp dst 202.164.30.1`

Display Filter Syntax 3

PART	EXAMPLE
Protocol field	`ip.addr`, `tcp.flags.syn`, `frame.len`
Comparison	`==`, `!=`, `>`, `<`, `>=`, `<=`
Value	Number, string, IP, or MAC address.
Logical	`and`, `or`, `not`, `xor`.

Syntax: `protocol.string comparison-operator value logical-operator expressions`

Example: `http.dest ip == 192.168.1.1 and tcp.port == 80`

Comparison Operators 4

OPERATOR	DESCRIPTION / EXAMPLE
`eq` / `==`	Equal — `ip.dest == 192.168.1.1`
`ne` / `!=`	Not equal — `ip.dest != 192.168.1.1`
`gt` / `>`	Greater than — `frame.len > 10`
`lt` / `<`	Less than — `frame.len < 10`
`ge` / `>=`	Greater than or equal — `frame.len >= 10`
`le` / `<=`	Less than or equal — `frame.len <= 10`

Logical & Special Operators 5

OPERATOR	DESCRIPTION
`and` / `&&`	Logical AND — all conditions must match.
`or` / `||`	Logical OR — any condition can match.
`xor` / `^^`	Logical XOR — only one of two conditions matches.
`not` / `!`	Negation — inverts the condition.
`[n]` / `[…]`	Substring operator — filter a specific word or text.
`{ }`	Membership operator — match a value in a set.

Slice operator `[...]` selects a range of values; the membership operator `{ }` checks "in" a list, e.g. `tcp.port in {80, 443, 8080}`.

Default Columns 6

COLUMN	MEANING
No.	Frame number from the beginning of the capture.
Time	Seconds from the first frame.
Source	Source address — commonly IPv4, IPv6 or Ethernet.
Destination	Destination address.
Protocol	Protocol used in the Ethernet frame, IP packet, or TCP segment.
Length	Length of the frame in bytes.

Miscellaneous — `Ctrl+E` starts/stops capturing.

Keyboard Shortcuts 7

SHORTCUT	ACTION
`Tab` / `Shift+Tab`	Move between screen elements (toolbars, packet list, detail).
`↓`	Move to the next packet or detail item.
`↑`	Move to the previous packet or detail item.
`Ctrl+↓` / `F8`	Next packet, even if the packet list isn't focused.
`Ctrl+↑` / `F7`	Previous packet, even if the packet list isn't focused.
`Ctrl+.`	Next packet of the conversation (TCP, UDP, or IP).
`Ctrl+,`	Previous packet of the conversation.
`Alt+→` / `Option+→`	Next packet in the selection history.
`→`	In packet detail, opens the selected tree item.
`Shift+→`	Open the tree item and all of its subtrees.
`Ctrl+→`	Open all tree items.
`Ctrl+←`	Close all tree items.
`Backspace`	Jump to the parent node.
`Return` / `Enter`	Toggle the selected tree item.
`Ctrl+E`	Start / stop capturing.

Main Toolbar 8

ITEM	MENU / DESCRIPTION
Start Capture	Capture → Start — same options as the previous session.
Stop Capture	Capture → Stop.
Restart Capture	Capture → Restart.
Options…	Capture → Options… — capture options dialog.
Open…	File → Open… — load a capture file.
Save As…	File → Save As… — save current capture.
Close	File → Close.
Reload	View → Reload — reload the current capture.
Find Packet…	Edit → Find Packet… — search by criteria.
Go Back / Forward	Go → Go Back / Go Forward — packet history.
Go to Packet…	Go → Go to Packet… — jump to a specific packet.
First / Last Packet	Go → First Packet / Last Packet.
Auto Scroll	View → Auto Scroll in Live Capture.
Colorize	View → Colorize — colorize the packet list.
Zoom In / Out / Normal	View → Zoom — adjust packet data font size.
Resize Columns	View → Resize Columns — fit content to width.

Common Display Filters 9

FILTER	EXAMPLE
Filter by IP	`ip.addr == 10.10.50.1`
Filter by source IP	`ip.src == 10.10.50.1`
Filter by destination IP	`ip.dest == 10.10.50.1`
Filter by IP range	`ip.addr >= 10.10.50.1 and ip.addr <= 10.10.50.100`
Filter by multiple IPs	`ip.addr == 10.10.50.1 and ip.addr == 10.10.50.100`
Filter out an IP	`!(ip.addr == 10.10.50.1)`
Filter a subnet	`ip.addr == 10.10.50.1/24`
Filter by port	`tcp.port == 25`
Filter by destination port	`tcp.dstport == 23`
IP address and port	`ip.addr == 10.10.50.1 and tcp.port == 25`
Filter by URL	`http.host == "hostname"`
Filter by timestamp	`frame.time >= "June 02, 2019 18:04:00"`
SYN flag	`tcp.flags.syn == 1`
SYN without ACK	`tcp.flags.syn == 1 and tcp.flags.ack == 0`
RST flag	`tcp.flags.reset == 1`
Beacon filter	`wlan.fc.type_subtype == 0x08`
Broadcast	`eth.dst == ff:ff:ff:ff:ff:ff`
Multicast	`(eth.dst[0] & 1)`
MAC address	`eth.addr == 00:70:f4:23:18:c4`
Host name	`ip.host == hostname`
Emails and IMF	`smtp.req.parameter == "MAIL FROM" OR imf.header == "From"`
DNS response code	`dns.resp.code == 3`